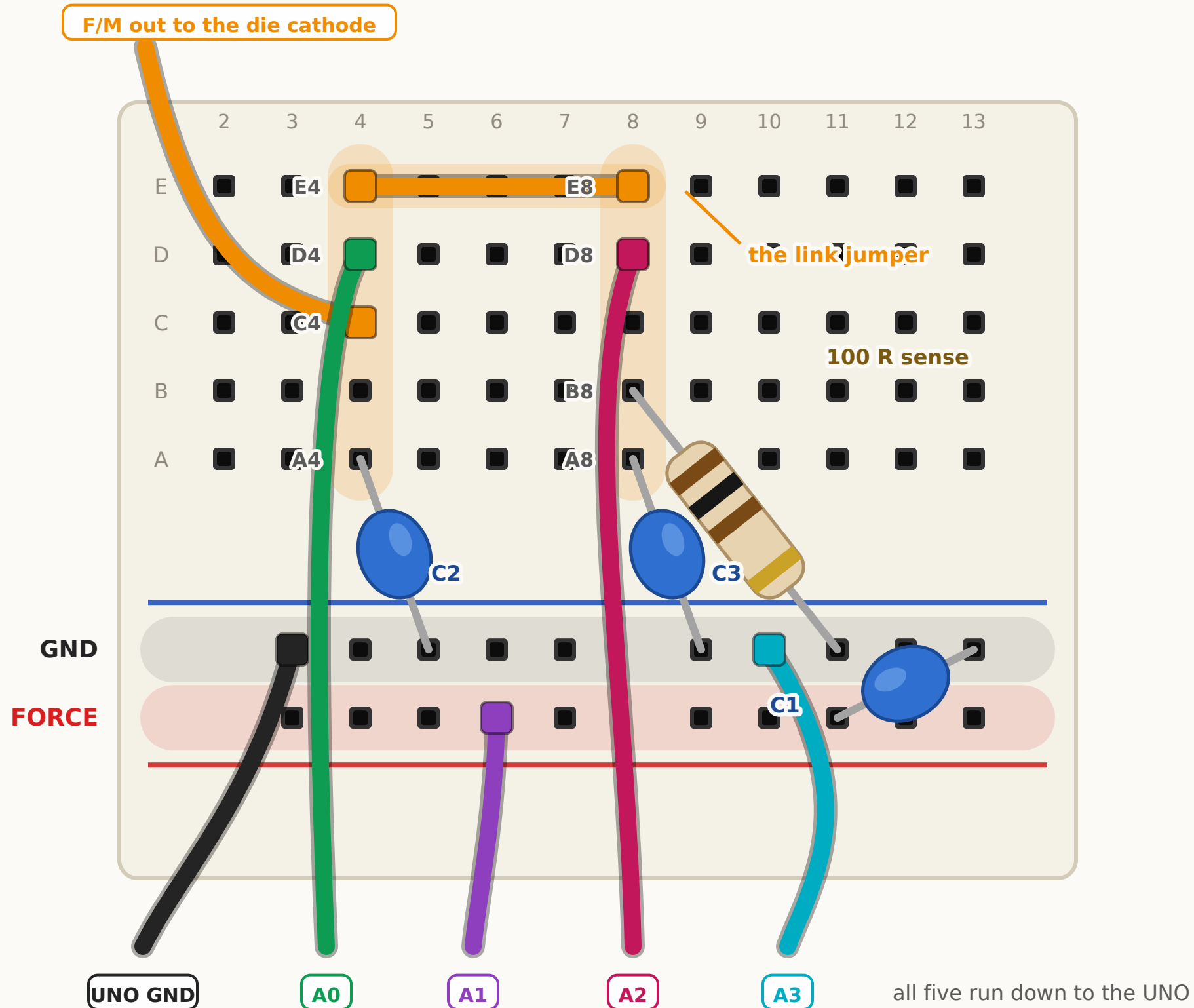


Close-up: the RETURN cluster and the two rails

Columns 2 to 13 at twice the size. This is the part that was crowded.



The orange shading is one single node

Holes A-E of one column are already joined inside the board. The link jumper joins column 4 to column 8, so all ten of those holes are RETURN. Anything you plug into any of them is on the same node.

Three things there, three different jobs

C4 - female-male jumper, leaves the board for the cathode pin of the channel under test
E4 to E8 - the link, lies flat on the board
D8 - goes to UNO A2, drawn pink so it cannot be confused with the two orange ones

How the rail holes are laid out

First rail hole is under column 3, then five in a row, a one-hole gap, five more, and so on. They line up with the numbered columns. The whole strip is one node, so the exact hole never changes the circuit.

What ends up on each rail

A3 taps the GND rail in the SAME five-hole group as the 100 R, so $i = (A2 - A3) / R_SENSE$ is a true difference
GND rail - UNO GND, A3, C1, C2, C3, the 100 R
FORCE rail - UNO A1, C1, the six bank resistors, and the female-male jumper to the die anode